

HandsOn-1

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User Stories

US-01: As a user, I want to log in with my username and password so that I can access the service.

US-02: As a user, I want to register with a username and password so that I can create an account.

US-03: As an administrator, I want to delete a user so that I can manage registered accounts.

US-04: As a user, I want to change my password so that I can keep my account secure.

US-05: As a user, I want to be locked out after multiple failed login attempts so that my account is protected from brute force attacks.

Definition of Done (DoD)

- **Code Quality:** Java code must compile without errors and follow standard naming conventions.
- **Functionality:** The feature must satisfy all requirements defined in the User Story.
- **Documentation:** Basic Javadoc comments must be included for all new classes and methods.

PBI-ID	Title / Description	Type	Status	Priority	Effort (SP)	State	Ready?
PBI-01	User login with username and password	Functionality	Ready for implementation	High	1	To do	Yes
PBI-02	User registration with username and password	Functionality	Ready for implementation	High	1	To do	Yes
PBI-03	Delete a user	Functionality	Ready for implementation	High	0.5	To do	Yes
PBI-04	Change user password	Functionality	Ready for implementation	Medium	0.5	To do	Yes
PBI-05	Re-engineer project from C to Java	Development activity	Ready for implementation	High	2	To do	Yes
PBI-06	Lock account after 3 failed login attempts	Functionality	Ready for refinement	Medium	1	To do	Yes
PBI-07	Rename existing user	Functionality	Ready for refinement	Low	0.5	To do	No

Sprint 1 Documentation

1. Sprint Goal

Establish the core Java project structure and implement the primary authentication features: User Registration and Login.

2. Sprint Backlog (Selected PBIs)

PBI ID	Description	Priority	Effort (SP)	Status
PBI-05	Re-engineer project structure from C to Java	High	2	Done
PBI-02	User registration functionality (REQ1)	High	1	Done
PBI-01	User login and credential verification (REQ2)	High	1	Done

3. Task Breakdown & Assignment

Task ID	Related PBI	Technical Task Description	Assigned To	Status
T1.1	PBI-05	Set up Maven/Java project and Git repository	Tuna	Done
T1.2	PBI-05	Create 'User' model class (username, password)	Tuna	Done
T2.1	PBI-02	Implement 'AuthService' with a HashMap for user storage	Tuna	Done
T2.2	PBI-02	Create 'registerUser' method with basic validation	Tuna	Done
T3.1	PBI-01	Create 'login' method to verify stored credentials	Tuna	Done

4. Daily Scrum Log (Simulation)

- **Day 1:** Set up the Java environment and initialized the Git repository. Created the base User class. No blockers.
- **Day 2:** Implemented the registerUser logic using a HashMap. Started working on the credential verification logic.
- **Day 3:** Completed the login method. Verified that registered users can successfully authenticate. All Sprint 1 tasks completed.

Sprint 2 Documentation

1. Sprint Goal

Implement administrative functionalities (User Deletion and Password Management) and enhance system security by introducing an account lockout mechanism after multiple failed login attempts.

2. Sprint Backlog (Selected PBIs)

PBI ID	Description	Priority	Effort (SP)	Status
PBI-03	Delete a user from the system (REQ1)	High	0.5	Done
PBI-04	Change existing user password (REQ1)	Medium	0.5	Done
PBI-06	Lock account after 3 failed login attempts (Security)	High	1	Done

3. Task Breakdown & Assignment

Task ID	Related PBI	Technical Task Description	Assigned To	Status
T3.1	PBI-03	Implement file rewriting logic to remove a user line	Tuna	Done
T4.1	PBI-04	Implement logic to update the password for a specific user	Tuna	Done
T6.1	PBI-06	Update 'User' class/storage to track failed login attempts	Tuna	Done
T6.2	PBI-06	Add lockout check logic to the 'login' method	Tuna	Done

4. Daily Scrum Log (Simulation)

- **Day 4:** Focused on administrative features. Successfully implemented the file rewriting logic to support `authDelete` and `authChangePassword`.
- **Day 5:** Addressed the "Brute Force" requirement. Modified the internal storage logic to keep track of failed attempts for each session.
- **Day 6:** Finalized the lockout mechanism. If a user fails 3 times, the `authCheck` method now returns a specific error/lockout status. Conducted final integration tests.